

WEEK 4 – ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

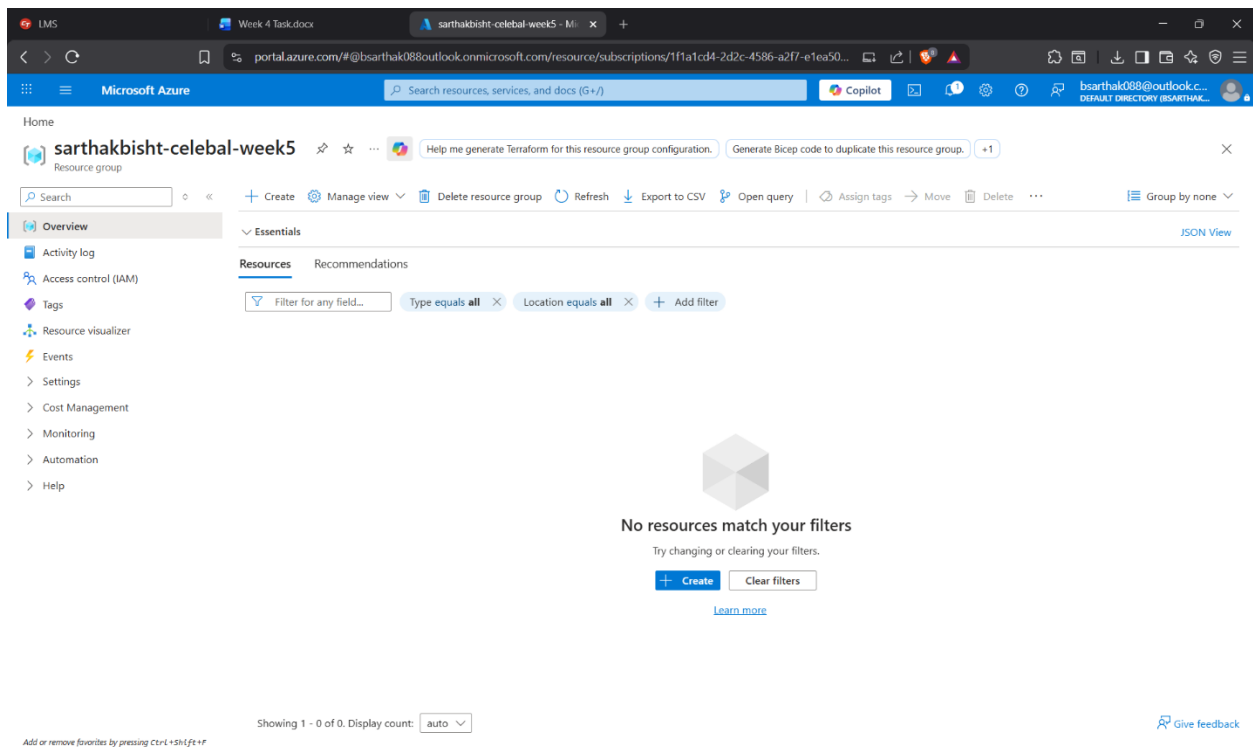
Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group

Screenshots: -



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Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

Screenshots: -

The screenshot shows the 'Create a storage account' page in the Microsoft Azure portal. The page is titled 'Create a storage account' and has a breadcrumb trail: 'Home > Storage center | Blob Storage'. The page is divided into several sections: 'Basics', 'Advanced', 'Networking', 'Data protection', 'Security', 'Encryption', 'Tags', and 'Review + create'. The 'Basics' section is active and contains the following information:

- Project details:** Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.
- Subscription:** Azure for Students
- Resource group:** sarthakbisht-celebal-week5
- Instance details:** Select the storage account name, region, and preferred storage type.
- Storage account name:** (Empty text box)
- Region:** (Asia Pacific) West India
- Preferred storage type:** Choose preferred storage type
- Performance:** Standard: Recommended for most scenarios (general-purpose v2 account)

At the bottom of the page, there are three buttons: 'Previous', 'Next', and 'Review + create'. The 'Review + create' button is highlighted in blue. In the bottom right corner, there is a link to 'Give feedback'.

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The screenshot shows the Microsoft Azure portal interface. The main heading is 'sarthakbisht1_1781430917026 | Overview'. The left sidebar contains 'Overview', 'Inputs', 'Outputs', and 'Template'. The main content area displays a green checkmark and the text 'Your deployment is complete'. Below this, deployment details are listed: Deployment name: sarthakbisht1_1781430917026, Subscription: Azure for Students, Resource group: sarthakbisht-celebal-week5, Start time: 6/14/2026, 6:14:36 PM, and Correlation ID: a518a0d2-e36f-4624-957e-eb99d84ddb2e. There are buttons for 'Delete', 'Cancel', 'Redeploy', 'Download', and 'Refresh'. Below the deployment details, there are sections for 'Deployment details' and 'Next steps' with a 'Go to resource' button. At the bottom, there are sections for 'Cost management' and 'Microsoft Defender for Cloud'.

Home

sarthakbisht1_1781430917026 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Your deployment is complete

Deployment name : sarthakbisht1_1781430917026
Subscription : Azure for Students
Resource group : sarthakbisht-celebal-week5
Start time : 6/14/2026, 6:14:36 PM
Correlation ID : a518a0d2-e36f-4624-957e-eb99d84ddb2e

Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Microsoft Defender for Cloud

Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

Add or remove favorites by pressing Ctrl+Shift+F

Notifications

More events in the activity log → Dismiss all

Deployment succeeded

Deployment 'sarthakbisht1_1781430917026' to resource group 'sarthakbisht-celebal-week5' was successful.
[Go to resource](#) [Go to resource group](#)
an hour ago

Resource group created

Creating resource group 'sarthakbisht-celebal-week5' in subscription 'Azure for Students' succeeded.
[Go to resource group](#) [Pin to dashboard](#)
4 hours ago

The screenshot shows the Microsoft Azure portal interface for the 'superstore-data' container. The main heading is 'superstore-data | Containers'. The left sidebar contains 'Overview', 'Diagnose and solve problems', 'Access Control (IAM)', and 'Settings'. The main content area displays the 'superstore-data' container details. It shows the authentication method as 'Access key (Switch to Microsoft Entra user account)'. There is a search bar for blobs and a table showing the contents of the container. The table has columns for Name, Last modified, Access tier, Blob type, and Size. One item is listed: 'Sample - Sup' with a size of 2.18 MiB. There are buttons for 'Add Directory', 'Upload', 'Change access level', 'Refresh', 'Delete', 'Copy', and 'Paste'. The right sidebar shows a 'Notifications' panel with three notifications: 'Successfully uploaded blob(s)', 'Successfully created storage container', and 'Deployment succeeded'.

Home > sarthakbisht1 | Containers

superstore-data

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste

superstore-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size
Sample - Sup	6/14/2026, 7:38:12 PM	Hot (Inferred)	Block blob	2.18 MiB

Notifications

More events in the activity log → Dismiss all

Successfully uploaded blob(s)

Successfully uploaded 1 blob(s).
a few seconds ago

Successfully created storage container

Successfully created storage container 'superstore-data'.
a few seconds ago

Deployment succeeded

Deployment 'sarthakbisht1_1781430917026' to resource group 'sarthakbisht-celebal-week5' was successful.
[Go to resource](#) [Go to resource group](#)
an hour ago

Resource group created

Creating resource group 'sarthakbisht-celebal-week5' in subscription 'Azure for Students' succeeded.
[Go to resource group](#) [Pin to dashboard](#)
4 hours ago

Add or remove favorites by pressing Ctrl+Shift+F

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Task 3: ADF Basics

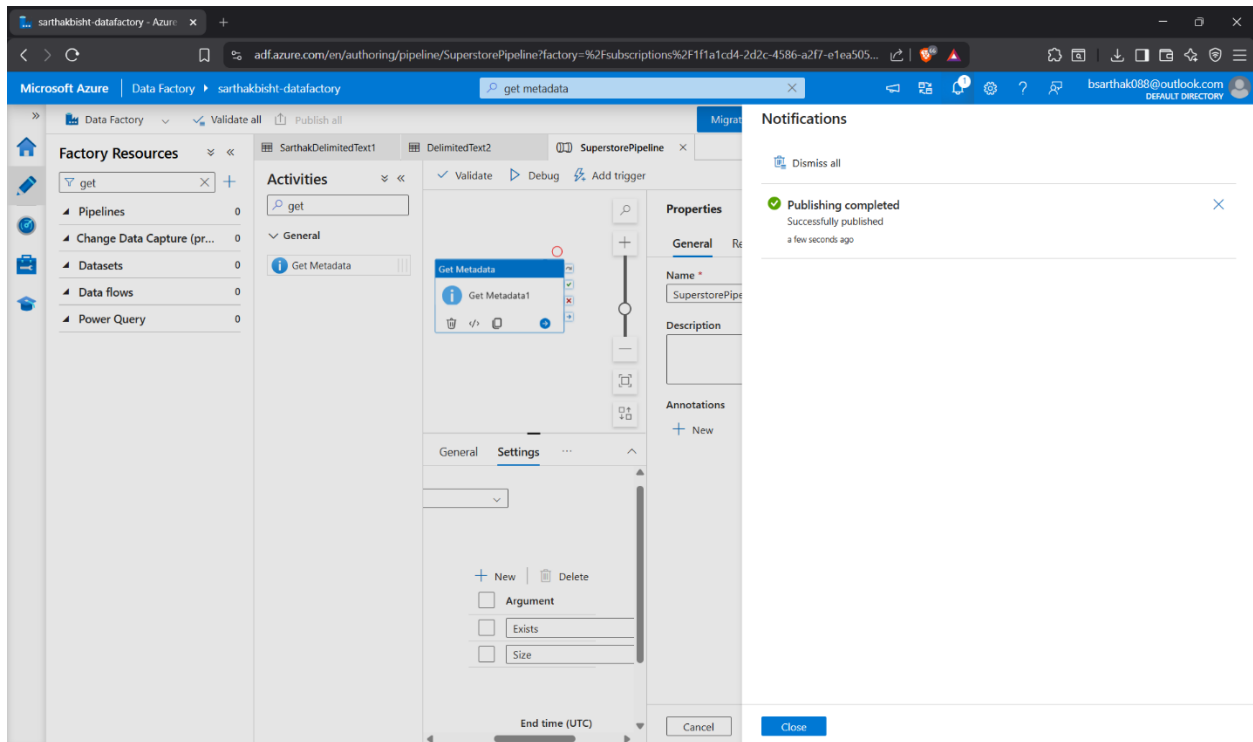
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

Screenshot: -

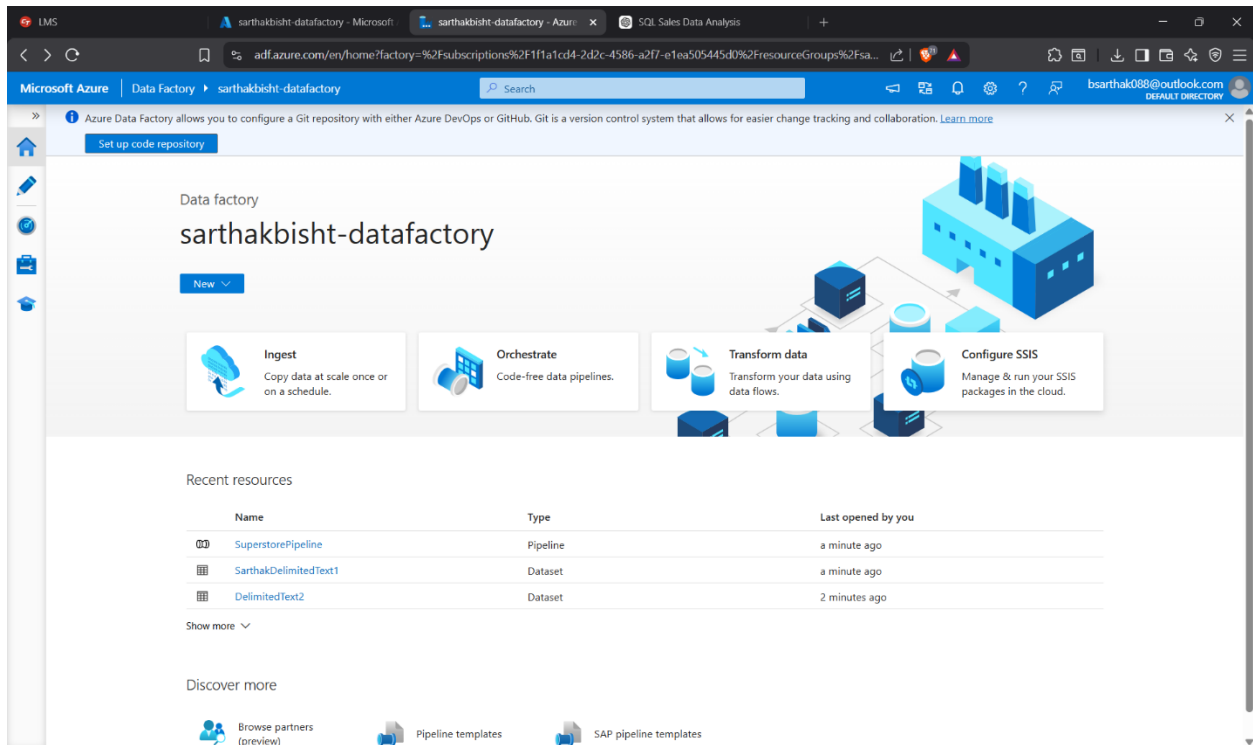


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The screenshot shows the Microsoft Azure Data Factory portal for the workspace 'sarthakbisht-datafactory'. The left sidebar displays 'Factory Resources' with categories like Pipelines, Change Data Capture, Datasets, Data flows, and Power Query. The main area is titled 'get metadata' and shows a 'DelimitedText' dataset named 'SarthakDelimitedText1'. The 'Properties' pane on the right includes fields for Name, Description, and Annotations. The 'Connection' pane shows the 'Linked service' as 'sarthakAzureBlobStorage1' and the 'File path' as 'superstore-data'. Other settings include 'Compression type' (No compression), 'Column delimiter' (Comma), 'Row delimiter' (Default), and 'Encoding' (Default(UTF-8)). A 'Pipeline validation output' pane on the right shows a success message: 'Your pipeline has been validated. No errors were found.'

The screenshot shows the 'Data factory sarthakbisht-datafactory' overview page. The page features a 'Set up code repository' button and a 'Now' dropdown. Below the header, there are four main sections: 'Ingest' (Copy data at scale once or on a schedule), 'Orchestrate' (Code-free data pipelines), 'Transform data' (Transform your data using data flows), and 'Configure SSIS' (Manage & run your SSIS packages in the cloud). The background includes a 3D illustration of a factory building.

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Task 4: Pipeline Development

Do:

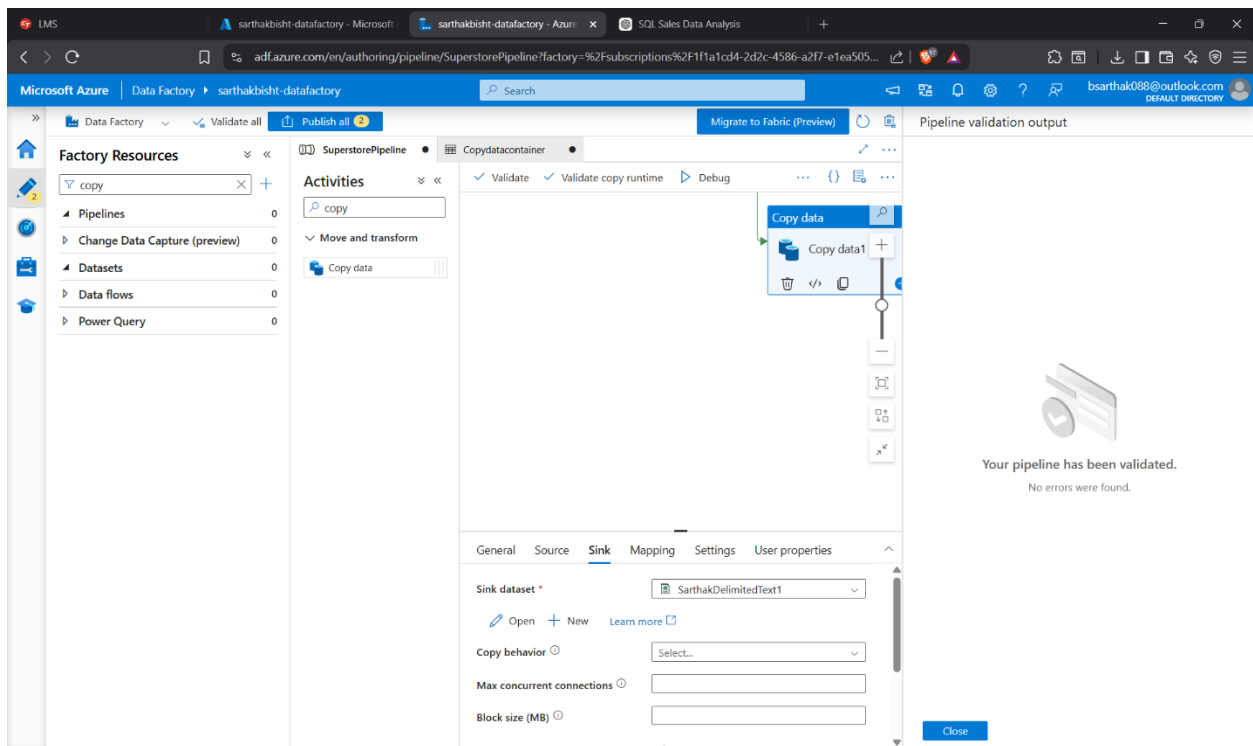
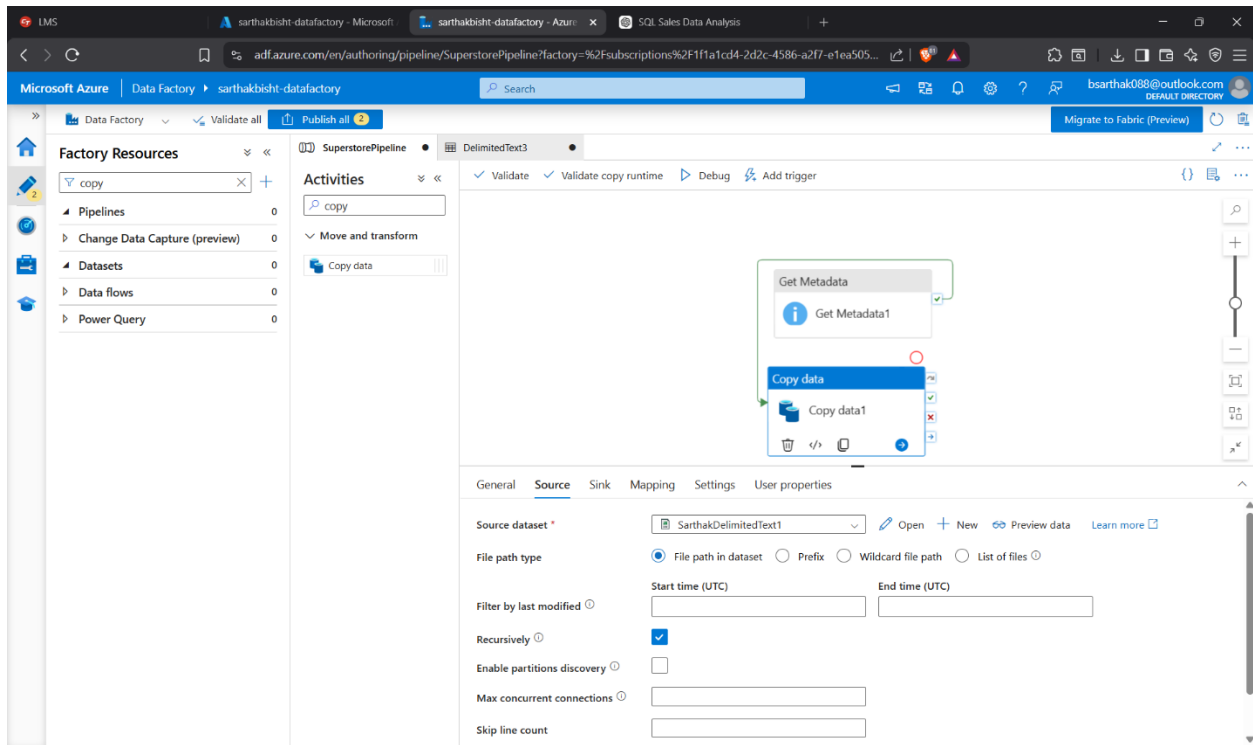
- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design

Screenshots: -

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Task 5: Pipeline Execution

Do:

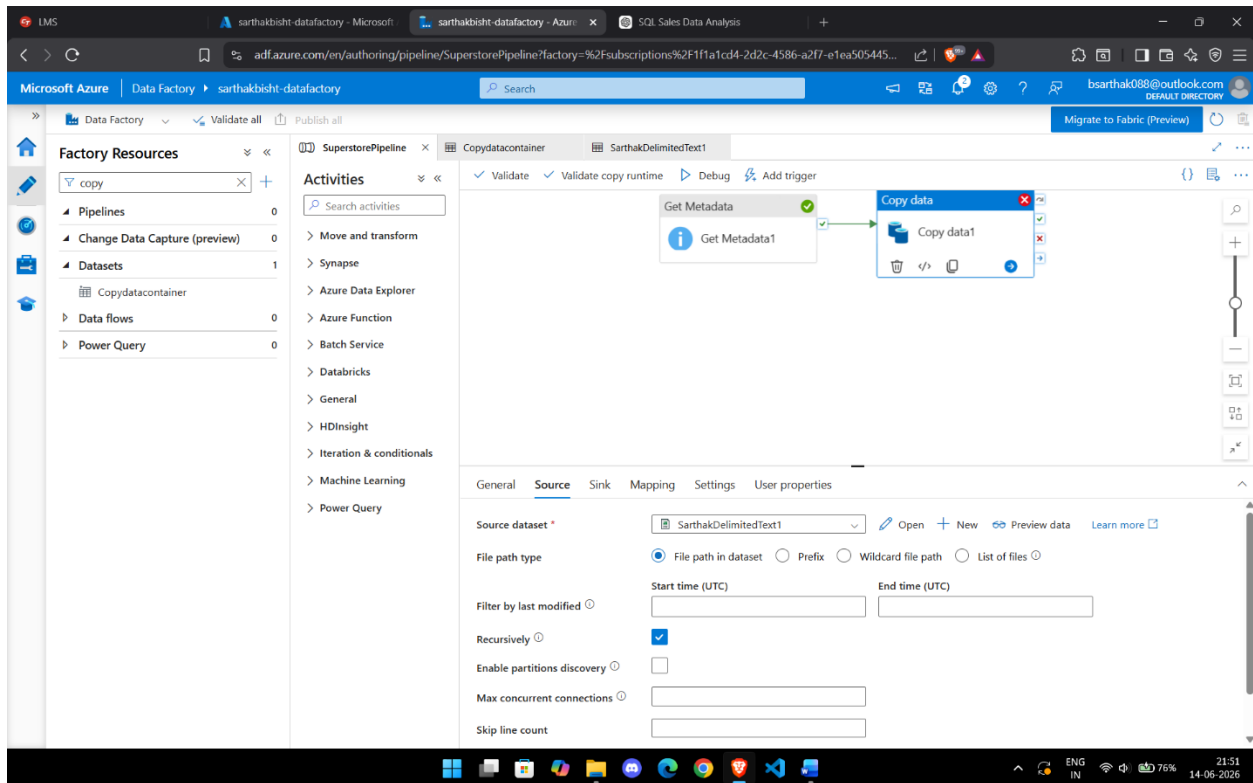
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- Run pipeline using Debug/Trigger

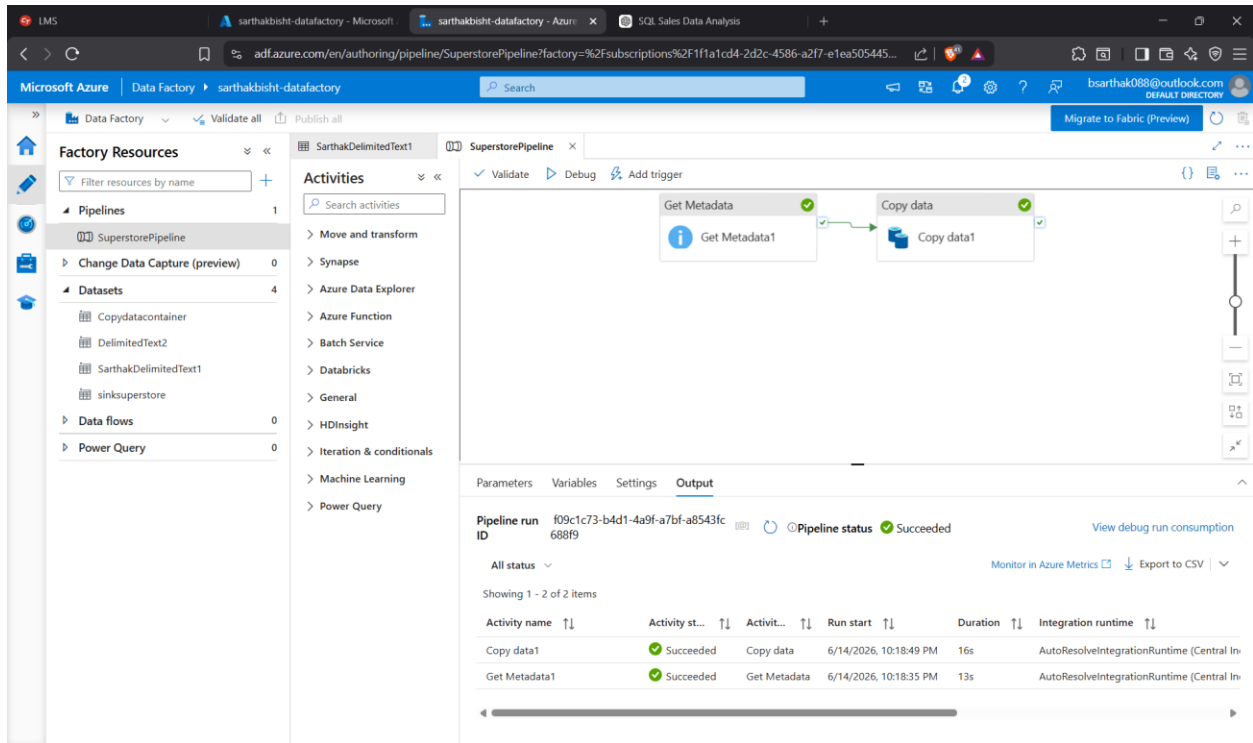
Deliverable:

- Screenshot showing pipeline execution (Succeeded)

Screenshots: -



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Task 6: IAM Roles

Do:

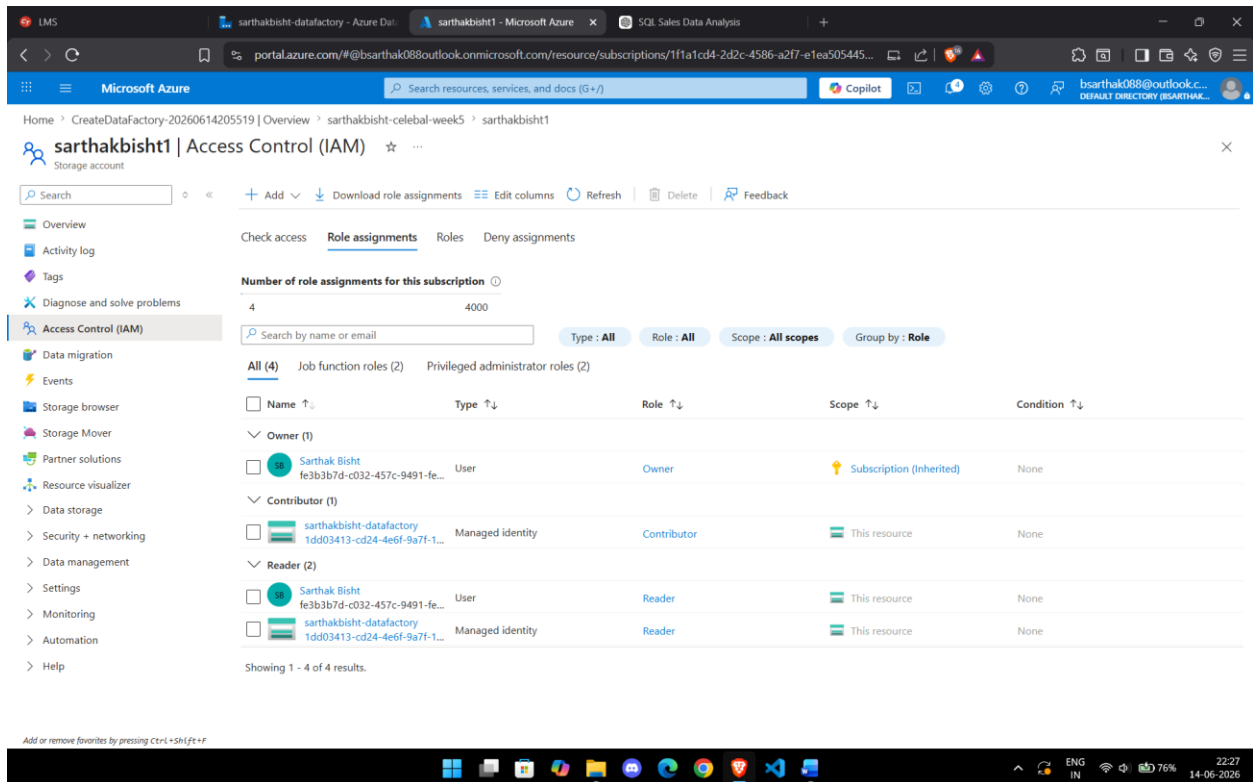
- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

Screenshots: -

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Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

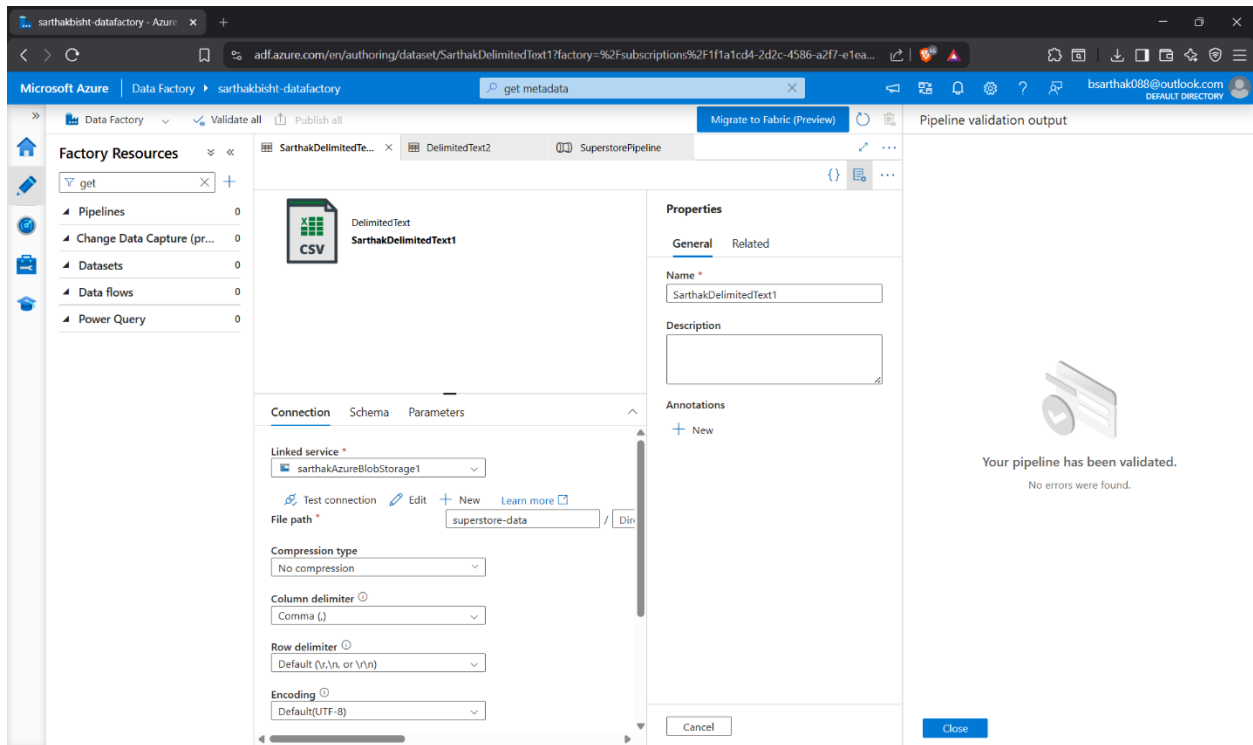
Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated

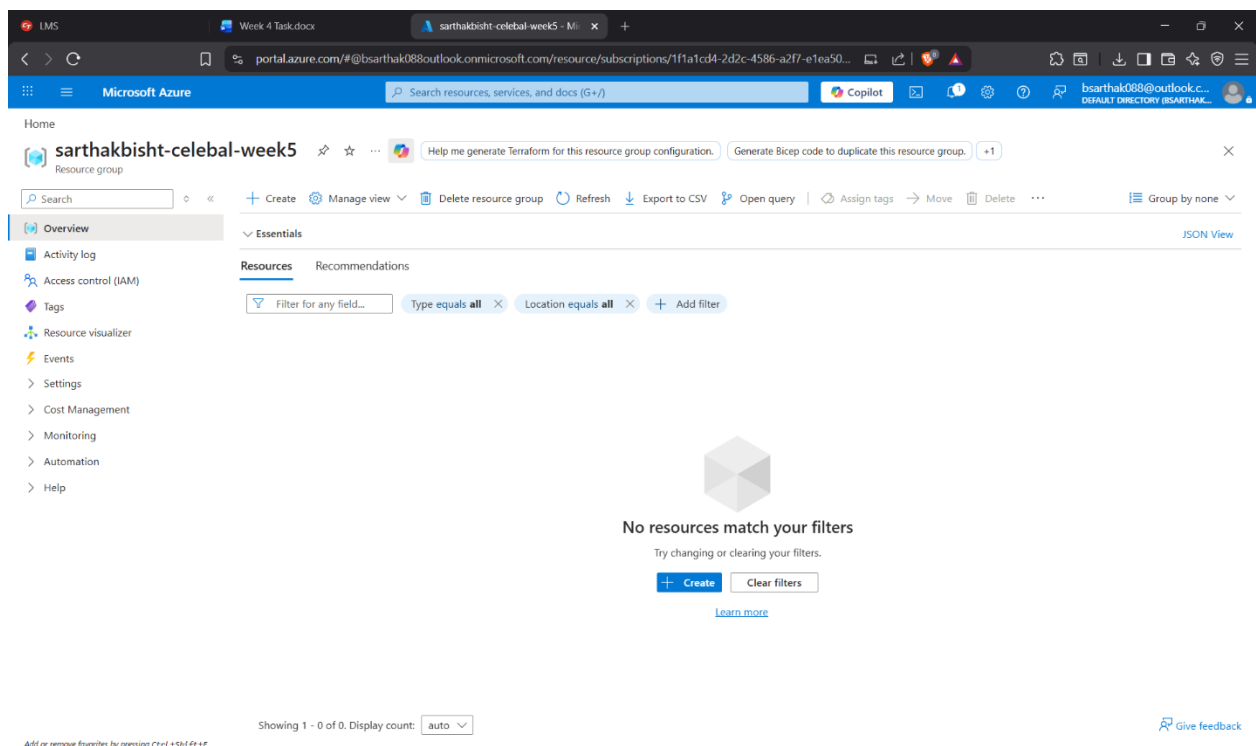
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validates the pipeline end-to-end execution along with some of the errors I encounter including :

*“ErrorCode=DelimitedTextMoreColumnsThanDefined
Error found when processing 'Csv/Tsv Format Text' source 'Sample - Superstore.csv'
with row number 34:
found more columns than expected column count 21“*

Debugged it using AI tools (ie. ChatGPT)



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The screenshot shows the Microsoft Azure portal interface. The main content area displays a deployment overview for the resource group 'sarthakbisht1_1781430917026'. The deployment is marked as 'Your deployment is complete' with a green checkmark. The deployment details show the deployment name, subscription, resource group, start time, and correlation ID. The next steps section includes a 'Go to resource' button. The cost management section provides information on staying within budget and preventing unexpected charges. The Microsoft Defender for Cloud section offers security for apps and infrastructure. The notifications panel on the right shows a list of events, including 'Deployment succeeded' and 'Resource group created', with links to go to the resource or resource group.

Home

sarthakbisht1_1781430917026 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : sarthakbisht1_1781430917026

Subscription : Azure for Students

Resource group : sarthakbisht-celebal-week5

Start time : 6/14/2026, 6:14:36 PM

Correlation ID : a518a0d2-e36f-4624-957e-eb99d94ddb2e

Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Add or remove favorites by pressing ctrl+shift+F

Notifications

More events in the activity log → Dismiss all

✓ Deployment succeeded

Deployment 'sarthakbisht1_1781430917026' to resource group 'sarthakbisht-celebal-week5' was successful.

Go to resource Go to resource group

an hour ago

✓ Resource group created

Creating resource group 'sarthakbisht-celebal-week5' in subscription 'Azure for Students' succeeded.

Go to resource group Pin to dashboard

4 hours ago

The screenshot shows the Microsoft Azure portal interface for a storage blob. The main content area displays the properties of the blob 'Sample - Superstore.csv'. The properties include URL, last modified, creation time, version ID, type, size, access tier, access tier last modified, archive status, rehydrate priority, server encrypted, etag, version-level immutability policy, cache-control, content-type, content-md5, content-encoding, content-language, content-disposition, lease status, lease state, and lease duration. The notifications panel on the right shows a list of events, including 'Successfully uploaded blob(s)', 'Successfully created storage container', and 'Deployment succeeded', with links to go to the resource or resource group.

Home

Sample - Superstore.csv

Blob

Save Discard Download Refresh Delete Change tier Acquire lease Break lease Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL https://sarthakbisht1.bl...

LAST MODIFIED 6/14/2026, 7:38:12 PM

CREATION TIME 6/14/2026, 7:38:12 PM

VERSION ID -

TYPE Block blob

SIZE 2.18 MiB

ACCESS TIER Hot (Inferred)

ACCESS TIER LAST MODIFIED N/A

ARCHIVE STATUS -

REHYDRATE PRIORITY -

SERVER ENCRYPTED true

ETAG 0x8DECA1E5EF160EF

VERSION-LEVEL IMMUTABILITY POLICY Disabled

CACHE-CONTROL

CONTENT-TYPE text/csv

CONTENT-MD5 swZpBentUdOU9Rz2/IMAg...

CONTENT-ENCODING

CONTENT-LANGUAGE

CONTENT-DISPOSITION

LEASE STATUS Unlocked

LEASE STATE Available

LEASE DURATION -

Notifications

More events in the activity log → Dismiss all

✓ Successfully uploaded blob(s)

Successfully uploaded 1 blob(s).

a few seconds ago

✓ Successfully created storage container

Successfully created storage container 'superstore-data'.

a few seconds ago

✓ Deployment succeeded

Deployment 'sarthakbisht1_1781430917026' to resource group 'sarthakbisht-celebal-week5' was successful.

Go to resource Go to resource group

an hour ago

✓ Resource group created

Creating resource group 'sarthakbisht-celebal-week5' in subscription 'Azure for Students' succeeded.

Go to resource group Pin to dashboard

4 hours ago

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This screenshot shows the Microsoft Azure Data Factory console. The left sidebar displays 'Factory Resources' with categories like Pipelines, Change Data Capture (preview), Datasets, Data flows, and Power Query. The main canvas shows a pipeline named 'SuperstorePipeline' with a single activity 'Copy data'. The 'Sink dataset' is set to 'SarthakDelimitedText1'. The right pane shows the 'Pipeline validation output' with a message: 'Your pipeline has been validated. No errors were found.'

This screenshot shows the 'Source' tab for the 'Copy data' activity in the 'SuperstorePipeline'. The 'Source dataset' is 'SarthakDelimitedText1'. The 'File path type' is set to 'File path in dataset'. The 'Filter by last modified' section includes 'Start time (UTC)' and 'End time (UTC)' fields. The 'Recursively' checkbox is checked. The 'Enable partitions discovery' checkbox is unchecked. The 'Max concurrent connections' and 'Skip line count' fields are empty.

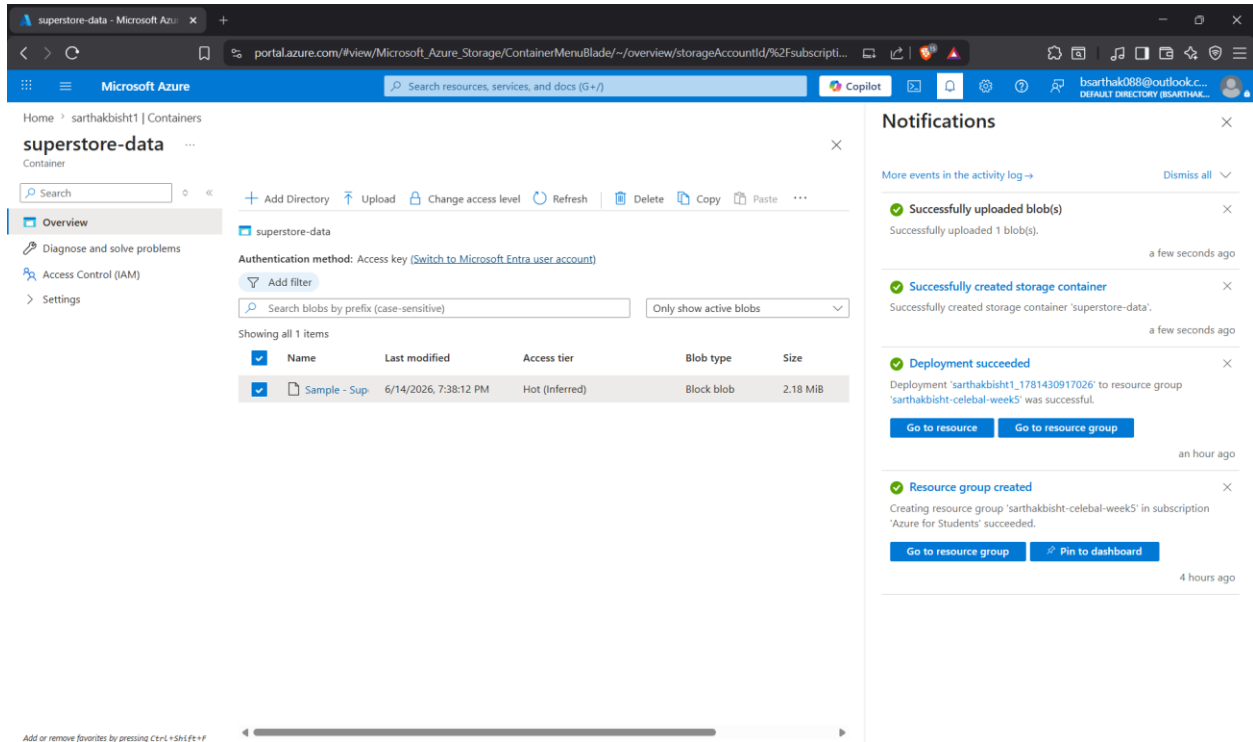
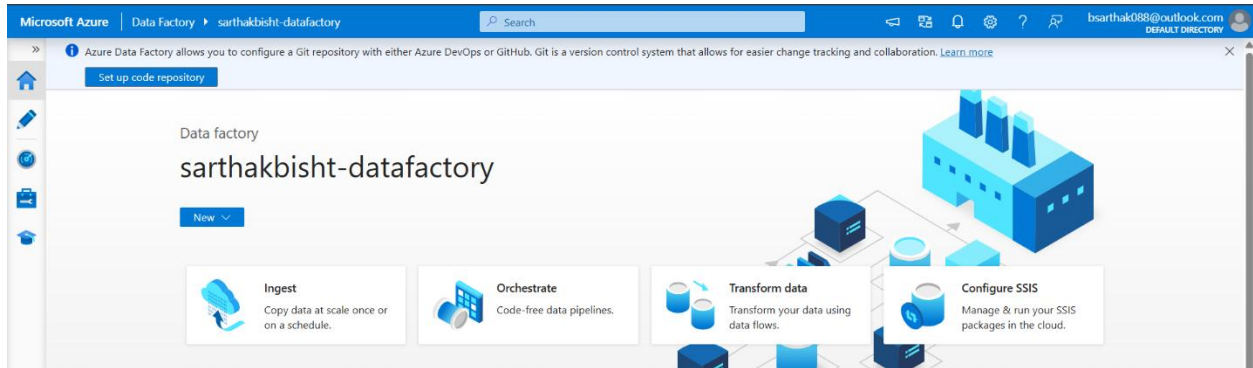
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The screenshot shows the Microsoft Azure Data Factory portal for the 'sarthakbisht-datafactory'. The page includes a 'Set up code repository' button, a 'Data factory sarthakbisht-datafactory' title, and a 'New' dropdown menu. Below these are four main action cards: 'Ingest' (Copy data at scale once or on a schedule), 'Orchestrate' (Code-free data pipelines), 'Transform data' (Transform your data using data flows), and 'Configure SSIS' (Manage & run your SSIS packages in the cloud). A 'Recent resources' table lists three items: 'SuperstorePipeline' (Pipeline, opened a minute ago), 'SarthakDelimitedText1' (Dataset, opened a minute ago), and 'DelimitedText2' (Dataset, opened 2 minutes ago). At the bottom, there are links to 'Browse partners (preview)', 'Pipeline templates', and 'SAP pipeline templates'.

Name	Type	Last opened by you
SuperstorePipeline	Pipeline	a minute ago
SarthakDelimitedText1	Dataset	a minute ago
DelimitedText2	Dataset	2 minutes ago

The screenshot shows the configuration page for the 'SarthakDelimitedText1' dataset in the 'sarthakbisht-datafactory'. The 'Factory Resources' sidebar on the left lists 'Pipelines' (0), 'Change Data Capture (pr...' (0), 'Datasets' (0), 'Data flows' (0), and 'Power Query' (0). The main area displays the dataset's 'Properties' (General and Related tabs) and 'Connection' details. The 'Connection' tab shows the 'Linked service' as 'sarthakAzureBlobStorage1' and the 'File path' as 'superstore-data'. Other settings include 'Compression type' (No compression), 'Column delimiter' (Comma), 'Row delimiter' (Default), and 'Encoding' (Default(UTF-8)). A 'Pipeline validation output' panel on the right shows a green checkmark and the message: 'Your pipeline has been validated. No errors were found.' Buttons for 'Cancel' and 'Close' are at the bottom.

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The image consists of two screenshots from the Microsoft Azure portal.

Top Screenshot: Access Control (IAM) Role assignments

The top screenshot shows the 'Access Control (IAM)' page for the subscription 'sarthakbisht1'. The 'Role assignments' tab is selected. It displays a table of role assignments for the subscription.

Name	Type	Role	Scope	Condition
Owner (1)				
Sarthak Bisht fe3b3b7d-c032-457c-9491-fe...	User	Owner	Subscription (Inherited)	None
Contributor (1)				
sarthakbisht-datafactory 1dd03413-cd24-4e6f-9a7f-1...	Managed identity	Contributor	This resource	None
Reader (2)				
Sarthak Bisht fe3b3b7d-c032-457c-9491-fe...	User	Reader	This resource	None
sarthakbisht-datafactory 1dd03413-cd24-4e6f-9a7f-1...	Managed identity	Reader	This resource	None

Showing 1 - 4 of 4 results.

Bottom Screenshot: Data Factory Pipeline Run

The bottom screenshot shows the 'Data Factory' page for the pipeline 'SuperstorePipeline'. The 'Output' tab is selected, displaying the results of the pipeline run.

Pipeline run ID: f09c1c73-b4d1-4a9f-a7bf-a8543fc688f9

Pipeline status: Succeeded

Activities:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	6/14/2026, 10:18:49 PM	16s	AutoResolveIntegrationRuntime (Central In
Get Metadata1	Succeeded	Get Metadata	6/14/2026, 10:18:35 PM	13s	AutoResolveIntegrationRuntime (Central In

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The image displays two screenshots of the Microsoft Azure Data Factory portal, illustrating the configuration and execution of a pipeline.

Top Screenshot: Pipeline Configuration

The interface shows the "SuperstorePipeline" with the "Copydatacontainer" activity selected. The "Source" tab is active, showing the "Source dataset" as "SarhakDelimitedText1". The "File path type" is set to "File path in dataset", and "Recursively" is checked. The "Sink" tab is also visible, showing the "Copy data" activity.

Bottom Screenshot: Pipeline Execution Status

The interface shows the "SuperstorePipeline" with the "Copydatacontainer" activity selected. The "Output" tab is active, showing the "Pipeline run ID" as "f64e80a0-f9c5-4436-9ccd-bfae3f468485". The "Status" is "All status". The "Showing 1 - 2 of 2 items" are listed:

Activity name	Activity st...
Copy data1	Failed
Get Metadata1	Succeeded

The "Notifications" panel on the right shows two "Publishing completed" messages, both "Successfully published".

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This screenshot shows the Microsoft Azure Data Factory portal. The left sidebar displays 'Factory Resources' with categories like Pipelines, Change Data Capture (preview), Datasets, Data flows, and Power Query. The main area shows a pipeline named 'SuperstorePipeline' with a 'Copydatacontainer' activity. The 'Activities' section lists 'copy' and 'Move and transform'. The 'Output' tab is selected, showing a table with columns: Pipeline run ID, Pipeline status, Activity name, Activity status, Activity, and Run start. The table shows a failed run for 'Copy data1' on 6/14/2026. The right sidebar displays 'Pipeline validation output' with a message: 'Your pipeline has been validated. No errors were found.'

Pipeline run ID	Pipeline status	Activity name	Activity status	Activity	Run start
f64e80a0-f9c5-4436-9c-cd-bfae3f468485	Failed	Copy data1	Failed	Copy data	6/14/2026

This screenshot shows the Microsoft Azure Data Factory portal, focusing on the 'Sink' tab for a 'Copy data' activity. The left sidebar is the same as the previous screenshot. The main area shows the 'Copy data' activity with a 'Copydatacontainer' activity. The 'Sink' tab is selected, showing a table with columns: Sink dataset, Copy behavior, Max concurrent connections, and Block size (MB). The table shows a sink dataset named 'SarhakDelimitedText1' with a copy behavior of 'Select...'. The right sidebar displays 'Pipeline validation output' with a message: 'Your pipeline has been validated. No errors were found.'

Sink dataset	Copy behavior	Max concurrent connections	Block size (MB)
SarakDelimitedText1	Select...		

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The image displays two screenshots of the Microsoft Azure Data Factory portal. The top screenshot shows the configuration of a pipeline named 'SuperstorePipeline'. The 'Activities' pane on the left lists 'copy' and 'Move and transform'. The 'Copy data' activity is selected, and its configuration is shown in the 'Source' tab. The 'Source dataset' is 'SarhakDelimitedText1'. The 'File path type' is 'File path in dataset'. The 'Filter by last modified' section is expanded, showing 'Start time (UTC)' and 'End time (UTC)' fields. The 'Recursively' checkbox is checked. The 'Enable partitions discovery' checkbox is unchecked. The 'Max concurrent connections' and 'Skip line count' fields are empty. The bottom screenshot shows the main dashboard of the 'sarhakbisht-datafactory'. The dashboard includes a 'Set up code repository' button, a 'New' button, and four main sections: 'Ingest' (Copy data at scale once or on a schedule), 'Orchestrate' (Code-free data pipelines), 'Transform data' (Transform your data using data flows), and 'Configure SSIS' (Manage & run your SSIS packages in the cloud). Below these sections is a 'Recent resources' table.

Name	Type	Last opened by you
SuperstorePipeline	Pipeline	a minute ago
SarthakDelimitedText1	Dataset	a minute ago
DelimitedText2	Dataset	2 minutes ago

Below the table is a 'Show more' link. At the bottom, there is a 'Discover more' section with three links: 'Browse partners (preview)', 'Pipeline templates', and 'SAP pipeline templates'.

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Microsoft Azure | Data Factory | sarthakbisht-datafactory

Set up code repository

Data factory

sarthakbisht-datafactory

New

Ingest
Copy data at scale once or on a schedule.

Orchestrate
Code-free data pipelines.

Transform data
Transform your data using data flows.

Configure SSIS
Manage & run your SSIS packages in the cloud.

Microsoft Azure | Data Factory | sarthakbisht-datafactory

get metadata

Validate all | Publish all | Migrate to Fabric (Preview) | Pipeline validation output

Factory Resources

- Pipelines: 0
- Change Data Capture (pr...): 0
- Datasets: 0
- Data flows: 0
- Power Query: 0

DelimitedText
SarhakDelimitedText1

Properties

General | Related

Name *
SarhakDelimitedText1

Description

Annotations
+ New

Connection | Schema | Parameters

Linked service *
sarthakAzureBlobStorage1

File path *
superstore-data

Compression type
No compression

Column delimiter
Comma (,)

Row delimiter
Default (\r\n, or \n\r)

Encoding
Default(UTF-8)

Cancel | Close

Your pipeline has been validated.
No errors were found.